

A Short Guide to LCCDEs

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1 Introduction

To be successful in finding the solution to a linear constant coefficient differential equation (LCCDE), one has to remember that:

1. The solution consists of two parts, a homogeneous solution plus a particular solution,

$$y(t) = y_h(t) + y_p(t)$$

2. The solution is *unique*, so that once a solution to the equation is found that satisfies the initial conditions (if any), then you are done.

2 Homogeneous Solution

For a linear constant coefficient differential equation of the form,

$$\sum_{n=0}^N a_n \frac{d^n y(t)}{dt^n} = \sum_{m=0}^M b_m \frac{d^m x(t)}{dt^m} \quad (1)$$

the homogeneous solution, $y_h(t)$, is one that satisfies the *homogeneous* equation, which is formed by setting the input $x(t)$ equal to zero,

$$\sum_{n=0}^N a_n \frac{d^n y_h(t)}{dt^n} = 0$$

Example 1: For the differential equation,

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = 5 \frac{d^2 x(t)}{dt^2} + 2x(t)$$

the homogeneous equation is

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = 0$$

and a homogeneous solution $y_h(t)$ is one that solves this equation.

Note that a homogeneous solution exists in the absence of any input, $x(t)$. Also note that for any solution $y_p(t)$ that solves the non-homogeneous differential equation, which is the one with $x(t)$ and derivatives of $x(t)$ on the right-hand side, then $y_p(t) + y_h(t)$ will also be a solution. Finally, note that for any input $x(t)$, once $x(t)$ goes to zero (the system is no longer excited by an input), then the only solution that persists is the homogeneous solution.

Homogeneous solutions are easy to find. They are linear combinations of functions (signals) of the form

$$y_h(t) = e^{st}$$

for certain values of s . The values of s that solve the homogeneous differential equation may be found by substituting e^{st} into the homogeneous equation as follows,

$$\sum_{n=0}^N a_n \frac{d^n}{dt^n} e^{st} = \sum_{n=0}^N a_n s^n e^{st} = e^{st} \left[\sum_{n=0}^N a_n s^n \right] = P(s) e^{st} = 0$$

where $P(s)$ is an N th-order polynomial in the variable s . Since $P(s)e^{st} = 0$ for all t , this can only be true if $P(s)$ is identically equal to zero,

$$P(s) = \sum_{n=0}^N a_n s^n = 0$$

This is called the *characteristic equation* of the LCCDE, and $P(s)$ is called the *characteristic polynomial*. Since $P(s)$ is an n th-order polynomial in s , then $P(s)$ has N roots, $s_1, s_2, \dots, s_N$, and we may write $P(s)$ in factored form as

$$P(s) = (s - s_1)(s - s_2) \cdots (s - s_N) = \prod_{n=1}^N (s - s_n) = 0$$

Therefore, it follows that any complex exponential of the form $e^{s_n t}$ is a solution to the homogeneous differential equation, where s_n is a root of $P(s)$. If the roots of $P(s)$ are distinct, i.e., $s_k \neq s_l$ for $k \neq l$, then the homogeneous solution has the form

$$y_h(t) = \sum_{n=1}^N A_n e^{s_n t}$$

Example 2: Consider the differential equation

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = 5 \frac{d^2 x(t)}{dt^2} + 2x(t)$$

In this case, the homogeneous equation is

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = 0$$

and the characteristic polynomial is

$$P(s) = s^2 + 3s + 2 = (s + 2)(s + 1)$$

Therefore, the homogeneous solution is

$$y_h(t) = A_1 e^{-2t} + A_2 e^{-t} \tag{2}$$

any A_1 and A_2 .

The analysis above holds when $P(s)$ has N distinct roots. If one or more of the roots of $P(s)$ are repeated (multiple-order roots), then the form of the homogeneous solution needs to be modified. For example, if $P(s)$ is a second-order polynomial with a second-order root at $s = s_0$,

$$P(s) = (s - s_0)^2$$

then the homogeneous solution will have the form

$$y_h(t) = (A_1 t + A_0) e^{s_0 t}$$

More generally, if $P(s)$ has a root of order r ,

$$P(s) = (s - s_0)^r Q(s)$$

where $Q(s)$ is some polynomial, then the homogeneous solution will contain terms of the form

$$y_h(t) = (A_{r-1} t^{r-1} + \cdots + A_1 t + A_0) e^{s_0 t}$$

in addition to those that come from the roots of $Q(s)$.

Example 3: For the differential equation

$$\frac{d^3y(t)}{dt^3} + 2\frac{d^2y(t)}{dt^2} + \frac{dy(t)}{dt} = \frac{dx(t)}{dt} - 4x(t)$$

the characteristic equation is

$$(s^3 + 2s^2 + s) = s(s^2 + 2s + 1) = s(s + 1)^2 = 0$$

Therefore, the roots of the characteristic equation are $s = 0$ along with a second-order root at $s = -1$, and so the homogeneous solution is

$$y_h(t) = (a_0 + a_1t)e^{-t} + c$$

for any a_0 and a_1 and for any constant c (note the constant c comes from the root $s = 0$, since e^{st} is equal to one when $s = 0$).

3 Particular Solution

When there is a non-zero input $x(t)$ driving the differential equation, it produces a particular solution $y_p(t)$ in addition to the homogeneous solution. Simply stated, the particular solution is one that solves the differential equation for the given input $x(t)$. Unfortunately, there are no general rules or procedures to find $y_p(t)$ for a general input. However, Table 1 provides a list of particular solutions for some of the most common inputs. The values of the constants in the particular solution will depend on the form of the differential equation.

Problem: Find the particular solution to the differential equation

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = 5\frac{d^2x(t)}{dt^2} + 2x(t)$$

when the input is a *ramp*,

$$x(t) = tu(t).$$

Solution

Since the input begins at time $t = 0$ and is given by $x(t) = t$ for $t \geq 0$, then from Table 1 we see that the particular solution has the form¹

$$y_p(t) = a_0 + a_1t \quad ; \quad t \geq 0$$

Substituting $x(t) = t$ and $y_p(t) = a_0 + a_1t$ into the differential equation, we have

$$3a_1 + 2(a_1t + a_0) = 2t$$

or

$$(2a_1 - 2)t + (3a_1 + a_0) = 0$$

Since this equation must hold for all values of time t , then it is clear that we must have

$$2a_1 - 2 = 0$$

$$3a_1 + a_0 = 0$$

Solving these two equations in two unknowns, we find that

$$a_0 = -\frac{3}{2}; \quad a_1 = 1$$

¹Note that since $x(t) = 0$ for $t < 0$, and since nothing is stated about any “initial conditions” that exist at time $t = 0$ or before, then $y(t) = 0$ for $t < 0$. More will be said about “initial conditions” in the next section.

$x(t)$	Particular Solution, $y_p(t)$
C	a_0
t	$a_1 t + a_0$
t^2	$a_2 t^2 + a_1 t + a_0$
$e^{\alpha t}$	$a_1 e^{\alpha t}$ if α is not a root of the characteristic equation. $(a_1 t + a_2) e^{\alpha t}$ if α is a root of the characteristic equation.
$\cos \omega_0 t$	$a_1 \cos \omega_0 t + a_2 \sin \omega_0 t$
$\sin \omega_0 t$	$a_1 \cos \omega_0 t + a_2 \sin \omega_0 t$

Table 1: Particular solutions to differential equations for some common inputs, $x(t)$.

Therefore, the particular solution is

$$y_p(t) = t - \frac{3}{2} \quad ; \quad t \geq 0$$

Recall that the homogeneous solution to this differential equation is given by Eq. (2), which is

$$y_h(t) = A_1 e^{-2t} + A_2 e^{-t}$$

Therefore, $y_h(t)$ for any constants A_1 and A_2 may be added to the particular solution, and it would still solve the differential equation. So, the total solution for $t \geq 0$ is

$$\begin{aligned} y(t) &= y_p(t) + y_h(t) \\ &= t - \frac{3}{2} + A_1 e^{-2t} + A_2 e^{-t} \quad ; \quad t \geq 0 \end{aligned}$$

Recall that the solution to a differential equation is unique. Therefore, although any values for A_1 and A_2 represent a solution to the differential equation, we have yet to define what the state of the system is at time $t = 0$ when the input is first applied. This is known as the set of *initial condition* that must be specified and, once this is done, the solution will be uniquely defined.

4 Solving LCCDEs with Initial Conditions

To find the complete solution to a LCCDE, one needs to specify a set of initial conditions. If the input starts at time $t = 0$, such as $x(t) = e^{-t}u(t)$, then these initial conditions are typically given in terms of the state of the system at time $t = 0$, i.e., just before the input is applied. In order to understand the underlying concept of *initial conditions*, note that for any system, such as a circuit or a mechanical system, if a current or voltage is applied to the circuit at time $t = 0$, or some force or torque is applied to a mechanical system at time $t = 0$, to find the total response of the system to this input, one needs to take into account what state the system was in at time $t = 0$. For example, there may be some initial charges stored on capacitors or some currents flowing through inductors when the input is applied, or in the case of a mechanical system, some masses may be in motion or rotating when a force is applied. These non-zero initial conditions may, for example, be the result of inputs that occurred prior to time $t = 0$ that are unknown but need to be accounted for.

For an N th order differential equation of the form given in Eq. (1), N initial conditions are required in order to determine the N constants $A_1, A_2, \dots, A_n$ in the homogeneous solution. Thus, the initial conditions may specify, for example, the value of $y(0)$ along with the values of the first $N - 1$ derivatives of $y(t)$ at $t = 0$. If the system is at initial rest, then these initial conditions would be

$$y(0) = 0 \quad \text{and} \quad \left. \frac{d^k y(t)}{dt^k} \right|_{t=0} = 0, \quad k = 1, 2, \dots, N - 1$$

Problem: Find the solution to the differential equation

$$\frac{d^2y(t)}{dt^2} + 3\frac{dy(t)}{dt} + 2y(t) = x(t)$$

for an input $x(t) = u(t)$ assuming that the system is in initial rest conditions at time $t = 0$, i.e.,

$$y(0) = y'(0) = 0$$

where we use $y'(0)$ to denote the derivative of $y(t)$ at time $t = 0$.

Solution

To solve this differential equation, we need to find the particular and homogeneous solutions, and then find the unknown constants in the homogeneous solution to satisfy the initial conditions. It does not matter whether one solves first for the particular solution and then the homogeneous, or vice versa. Here we begin with the particular solution.

Since the input has a constant value of one for $t \geq 0$, then from Table 1 it follows that the particular solution for $t \geq 0$ is a constant,

$$y_p(t) = a_0$$

Substituting $x(t) = 1$ and $y_p(t) = a_0$ into the differential equation, which is valid for all $t \geq 0$, we have,²

$$2a_0 = 1$$

or $a_0 = 1/2$ and

$$y_p(t) = \frac{1}{2}$$

The homogeneous solution to this differential equation is given by Eq. (2), and so the total solution is

$$y(t) = \frac{1}{2} + A_1e^{-2t} + A_2e^{-t}$$

The constants A_1 and A_2 must now be found so that the total solution satisfies the initial rest conditions

$$y(0) = y'(0) = 0$$

The first initial condition gives

$$y(0) = \frac{1}{2} + A_1 + A_2 = 0$$

whereas from the second we have

$$y'(0) = -2A_1 - A_2 = 0$$

Solving these two linear equations in two unknowns, we find that

$$A_1 = 1/2 \quad ; \quad A_2 = -1$$

Therefore, the total solution, for $t \geq 0$ is³

$$y(t) = \frac{1}{2} + \frac{1}{2}e^{-2t} - e^{-t}$$

or

$$y(t) = \left[\frac{1}{2} + \frac{1}{2}e^{-2t} - e^{-t} \right] u(t)$$

In the previous problem, the step response of the linear time-invariant system defined by the given differential equation was found. Since the impulse response is the derivative of the step response, then the impulse response is

$$h(t) = \frac{dy(t)}{dt} = \frac{d}{dt} \left\{ \left[\frac{1}{2} + \frac{1}{2}e^{-2t} - e^{-t} \right] u(t) \right\}$$

²Note that since $y_p(t)$ is equal to a constant for time $t > 0$, then the derivatives on the left-hand side of the differential equation are equal to zero.

³One may legitimately ask what the solution is for time $t < 0$, but generally this is not of any interest. We are given the state of the system at time $t = 0$, and what we would like to determine is the output $y(t)$ for all $t \geq 0$. How the system got into its initial state is not important.

Here we need to be a little careful in computing the derivatives because there may be a discontinuity in $y(t)$ at time $t = 0$. In this case, however, since $y(0) = 0$, then $y(t)$ is continuous at $t = 0$ and the derivative is simply

$$h(t) = (e^{-t} - e^{-2t}) u(t)$$

Problem: Find the response of the linear time-invariant system described by the LCCDE

$$\frac{dy(t)}{dt} + y(t) = x(t)$$

where $x(t) = \cos(t)$ and $x(0) = 1$.

Solution

For this differential equation, the characteristic polynomial is

$$P(s) = s + 1$$

and the homogeneous solution is

$$y_h(t) = Ae^{-t}$$

Since the input is a cosine, from Table 1 we see that the particular solution is

$$y_p(t) = a_1 \cos(t) + a_2 \sin(t)$$

Substituting the expression for $y_p(t)$ into the differential equation along with $x(t) = \cos(t)$, we have

$$-a_1 \sin(t) + a_2 \cos(t) + a_1 \cos(t) + a_2 \sin(t) = \cos(t)$$

or

$$(a_1 + a_2 - 1) \cos(t) + (a_2 - a_1) \sin(t) = \cos(t)$$

Since this must hold for all $t \geq 0$, then the coefficients multiplying the sine and the cosine must be equal to zero,

$$a_1 + a_2 - 1 = 0 \quad \text{and} \quad a_2 - a_1 = 0$$

Solving for a_1 and a_2 we find that

$$a_1 = a_2 = \frac{1}{2}$$

Therefore, the particular solution for $t \geq 0$ is

$$y_p(t) = \frac{1}{2} \cos(t) + \frac{1}{2} \sin(t)$$

Putting everything together, the total solution is

$$y_t(t) = y_h(t) + y_p(t) = Ae^{-t} + \frac{1}{2} \cos(t) + \frac{1}{2} \sin(t) \quad ; \quad t \geq 0$$

To solve for the constant A , we use the given initial condition, $y(0) = 0$, which gives

$$A + \frac{1}{2} = 0 \longrightarrow A = -\frac{1}{2}$$

and

$$y_t(t) = \frac{1}{2} (-e^{-t} + \cos(t) + \sin(t)) u(t)$$

The following problem shows how to find the impulse response of an LTI system described by a differential equation. Instead of first finding the step response and then taking the derivative, $h(t)$ is found directly. In order to do so, it is necessary to *balance* the singularities on both sides of the equation.

Problem: Find the impulse response of the linear time-invariant system described by the LCCDE

$$\frac{d^2 y(t)}{dt^2} + 2 \frac{dy(t)}{dt} + y(t) = x(t)$$

assuming initial rest conditions.

Solution

One approach to solve this equation would be to find the step response, and then differentiate the step response to find the impulse response. However, we may also find the impulse response directly.

The characteristic equation of this differential equation is

$$s^2 + 2s + 1 = (s + 1)^2 = 0$$

Since there is a second-order root at $s = -1$, then the homogeneous solution is

$$y_h(t) = (A_0 + A_1 t)e^{-t}$$

Since the system is at initial rest at time $t = 0^-$, just before the impulse occurs, and since $x(t) = 0$ for $t < 0$, then $y(t) = 0$ for $t < 0$.

For $t > 0$ the input is zero so the particular solution is also zero, and the total solution for $t > 0$ is

$$y(t) = (A_0 + A_1 t)e^{-t}$$

Although the system is in initial rest at time $t = 0^-$, the impulse will produce a discontinuity in the initial rest conditions. More specifically, note that since there is an impulse $\delta(t)$ on the right side of the differential equation, there must be an impulse on the left side. The impulse must come from the second derivative of $y(t)$,

$$\frac{d^2 y(t)}{dt^2} = \delta(t)$$

and so there is a step discontinuity in the first derivative of $y(t)$ of amplitude one,

$$\frac{dy(t)}{dt} = u(t)$$

Therefore, with $y'(0^-) = 0$ it follows that $y'(0^+) = 1$. With this initial condition along with $y(0^+) = 0$, we have

$$y(0^+) = A_0 = 0$$

and with

$$y'(t) = A_1 e^{-t} - (A_0 + A_1 t)e^{-t}$$

we have

$$y'(0^+) = A_1 - A_0 = 1 \longrightarrow A_1 = 1$$

Therefore, the solution to the differential equation is

$$y(t) = te^{-t}u(t)$$

The previous problem may cause some confusion in what is happening when we are finding the impulse response since the zero initial conditions seem to have magically, and without reason, been changed. The way to understand what happens when the input on the right side of the equation is an impulse is that this impulse produces a (very short) response of the system to this input, changing the initial zero state into one that is non-zero. Consider, for example, the case where the input is a very narrow pulse of duration ϵ and of amplitude $1/\epsilon$. Over this very short period of time of duration ϵ , the system produces a non-zero response, and at time $t = \epsilon$ the output $y(t)$ will be non-zero with, in general, non-zero derivatives. For $t > \epsilon$, the input goes to zero and the only remaining response is the homogeneous solution. As we let $\epsilon \rightarrow 0$, this narrow pulse becomes an impulse, and time $t = \epsilon$ becomes time $t = 0^+$, and the state of the system at time $t = 0^+$ is non-zero.